

```
In [1]: import pandas as pd

df = pd.read_csv("/home/hanysaa/truck.csv")

df.head()
```

```
Out[1]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>X</th>
<th>Y</th> </thead> <tbody> <th>0</th> <th>1</th>
<th>2</th> <th>3</th> <th>4</th> </tbody> <tbody>
</tbody>
```

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12.4  11.2
```

```
14.3  12.5
```

```
14.5  12.7
```

```
14.9  13.1
```

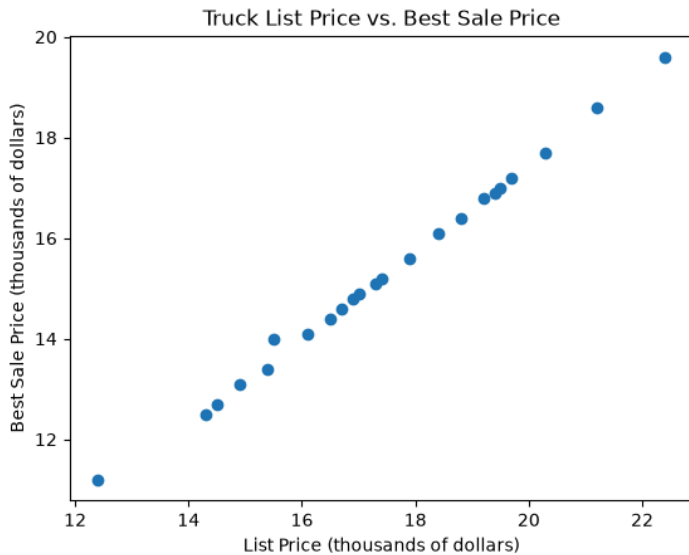
```
16.1  14.1
```

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In [2]: df["X"].corr(df["Y"])
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Out[2]: np.float64(0.9985404972960461)
```

```
In [3]: import matplotlib.pyplot as plt

plt.scatter(df["X"], df["Y"])
plt.xlabel("List Price (thousands of dolla
plt.ylabel("Best Sale Price (thousands of
plt.title("Truck List Price vs. Best Sale
plt.show()
```



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In [4]: df["X"].min(), df["X"].max()
```

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Out[4]: (np.float64(12.39999962), np.float64(22.39999962))
```

```
In [5]: import numpy as np

slope, intercept = np.polyfit(df["X"], df["Y"])

list_price = 17.40
predicted_best_price = slope * list_price + intercept

predicted_best_price
```

```
Out[5]: np.float64(15.244490749656322)
```

<p>My confidence in the prediction is very high because the Pearson correlation equal 0.9985 which is very close to 1 .which represent strong linear relationship between list price and sale price.I choose a list price of 17400 and i got a 15244 as a

best price which both results within the range .so
the prediction is supported the observation
data.however.the sale price is estimated and not
the actual sale price.</p>

In []: